

Міністерство освіти і науки України
Національний технічний університет України
«Київський політехнічний інститут імені Ігоря Сікорського»
Фізико-технічний інститут

Проектування розподілених систем

Лабораторна робота 5
«Мікросервіси з використанням Service Discovery та Config Server на базі Consul»

Виконав:
Студент групи ФБ-42мп
Осіпчук Антон

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До системи необхідно додати *Consul*, який буде виконувати роль *Service Register*, *Service Discovery* та *Config Server*.

Це дасть змогу явно не вказувати адреси мікросервісів та конфігурації для Hazelcast та Message Queue

1. Всі мікросервіси мають самостійно динамічно реєструватись при старті у *Consul*, кожного з сервісів може бути запущено кілька екземплярів (це буде відображатись як кількість instances на UI):

- **facade-service**
- **logging-service**
- **messages-service**

```
logging-service3-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:30] "GET /health HTTP/1.1" 200 -
consul-1 | 2025-06-17T17:19:30.989Z [INFO] agent: Synced check: check=service:log3
messages-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:33] "GET /health HTTP/1.1" 200 -
facade-service-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:36] "GET /health HTTP/1.1" 200 -
logging-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:38] "GET /health HTTP/1.1" 200 -
logging-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:38] "GET /health HTTP/1.1" 200 -
messages-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:39] "GET /health HTTP/1.1" 200 -
logging-service3-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:40] "GET /health HTTP/1.1" 200 -
messages-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:43] "GET /health HTTP/1.1" 200 -
facade-service-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:46] "GET /health HTTP/1.1" 200 -
logging-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:48] "GET /health HTTP/1.1" 200 -
logging-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:48] "GET /health HTTP/1.1" 200 -
messages-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:49] "GET /health HTTP/1.1" 200 -
logging-service3-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:50] "GET /health HTTP/1.1" 200 -
messages-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:53] "GET /health HTTP/1.1" 200 -
facade-service-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:56] "GET /health HTTP/1.1" 200 -
logging-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:58] "GET /health HTTP/1.1" 200 -
logging-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:58] "GET /health HTTP/1.1" 200 -
messages-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:19:59] "GET /health HTTP/1.1" 200 -
logging-service3-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:20:00] "GET /health HTTP/1.1" 200 -
messages-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:20:03] "GET /health HTTP/1.1" 200 -
```

Services 4 total

Q Search	Search Across	Health Status	Service Type	Unhealthy to Healthy
consul		1 instance		
facade-service		1 instance		
messages-service		2 instances		
logging-service		3 instances		

При звертанні **facade-service** до **logging-service** та **messages-service**, IP-адреси (і порти) мають читуватись **facade-service** з *Consul*. Немає бути задано в коді чи конфігураціях статичних значень адрес.

Відправка повідомлення через facade-service (POST /msg)

```
D:\dist-sys\Task5>curl -X POST http://localhost:8880/msg -H "Content-Type: application/json" -d '{"content": "Test message"}' {"status": "Sent to Kafka and logging-service"}
```

```
D:\dist-sys\Task5>
```

```
facade-service-1 | INFO:kafka.conn:<BrokerConnection client_id=kafka-python-producer-2, node_id=2 host=kafka2:9093 <connected> [IPv4 ('172.19.0.8', 9093
)]>: Connection complete.
facade-service-1 | INFO:kafka.conn:<BrokerConnection client_id=kafka-python-producer-2, node_id=bootstrap-2 host=kafka1:9092 <connected> [IPv4 ('172.19.
0.9', 9092)]>: Closing connection.
facade-service-1 | INFO:__main__: [facade1] Message sent to Kafka: Test message
facade-service-1 | INFO:__main__: [facade1] Fetched URLs for logging-service: ['http://logging-service1:8881', 'http://logging-service2:8882', 'http://lo
gging-service3:8883']
logging-service1-1 | INFO:__main__: [log1] Stored message: Test message
logging-service1-1 | INFO:werkzeug:172.19.0.17 - - [17/Jun/2025 17:22:33] "POST /Log HTTP/1.1" 201 -
facade-service-1 | INFO:__main__: [facade1] Successfully sent to http://logging-service1:8881
facade-service-1 | INFO:__main__: [facade1] Successfully sent to logging-service: {'status': 'Stored'}
facade-service-1 | INFO:werkzeug:172.19.0.1 - - [17/Jun/2025 17:22:33] "POST /msg HTTP/1.1" 201 -
messages-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:22:33] "GET /health HTTP/1.1" 200 -
```

```

messages-service1-1 | INFO:kafka.consumer.fetcher:Resetting offset for partition TopicPartition(topic='msg-queue', partition=0) to offset 0.
messages-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:24:21] "GET /health HTTP/1.1" 200 -
logging-service3-1 |

```

Logs by messages-service1:

```

9093 [C 172.19.0.8 , 9093] IPv4]
INFO:kafka.conn:<BrokerConnection client_id=kafka-python-2.2.11, node_id=2 host=kafka2:9093
INFO:kafka.consumer.fetcher:Resetting offset for partition TopicPartition(topic='msg-queue',
INFO:__main__:[msg1] Consumed message: Test message
INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:24:23] "GET /health HTTP/1.1" 200 -
INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:24:33] "GET /health HTTP/1.1" 200 -
INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:24:43] "GET /health HTTP/1.1" 200 -
INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:24:53] "GET /health HTTP/1.1" 200 -
INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:25:03] "GET /health HTTP/1.1" 200 -
INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:25:13] "GET /health HTTP/1.1" 200 -
INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:25:23] "GET /health HTTP/1.1" 200 -
INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:25:33] "GET /health HTTP/1.1" 200 -
INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:25:43] "GET /health HTTP/1.1" 200 -
PS D:\dist-sys\Task5>

```

Отримання комбінованих повідомлень (GET /msg)

```

logging-service3-1 | INFO:__main__:[log3] Fetched 1 messages
logging-service3-1 | INFO:werkzeug:172.19.0.17 - - [17/Jun/2025 17:26:55] "GET /log HTTP/1.1" 200 -
facade-service1 | INFO:__main__:[facade1] Successfully sent to http://logging-service3:8883
messages-service2-1 | INFO:__main__:[msg2] Returning 0 messages
facade-service1 | INFO:__main__:[facade1] Fetched URLs for messages-service: ['http://messages-service1:8890', 'http://messages-service2:8891']
messages-service2-1 | INFO:werkzeug:172.19.0.17 - - [17/Jun/2025 17:26:55] "GET /msg HTTP/1.1" 200 -
facade-service1 | INFO:__main__:[facade1] Fetched URLs for messages-service: ['http://messages-service1:8890', 'http://messages-service2:8891']
facade-service1 | INFO:__main__:[facade1] Successfully sent to http://messages-service2:8891
facade-service1 | INFO:werkzeug:172.19.0.1 - - [17/Jun/2025 17:26:55] "GET /msg HTTP/1.1" 200 -
facade-service1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:26:57] "GET /health HTTP/1.1" 200 -
logging-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:26:58] "GET /health HTTP/1.1" 200 -
logging-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:26:59] "GET /health HTTP/1.1" 200 -
messages-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:26:59] "GET /health HTTP/1.1" 200 -
logging-service3-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:01] "GET /health HTTP/1.1" 200 -
messages-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:03] "GET /health HTTP/1.1" 200 -
facade-service1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:07] "GET /health HTTP/1.1" 200 -
logging-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:08] "GET /health HTTP/1.1" 200 -
logging-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:09] "GET /health HTTP/1.1" 200 -
messages-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:09] "GET /health HTTP/1.1" 200 -
logging-service3-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:11] "GET /health HTTP/1.1" 200 -
messages-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:13] "GET /health HTTP/1.1" 200 -
facade-service1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:17] "GET /health HTTP/1.1" 200 -
logging-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:18] "GET /health HTTP/1.1" 200 -
logging-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:19] "GET /health HTTP/1.1" 200 -
messages-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:27:19] "GET /health HTTP/1.1" 200 -

```

```

D:\dist-sys\Task5>curl -X GET http://localhost:8880/msg
{"combined_messages":[{"content":"Test message (handled by log1)"}],"log_services":[{"messages":["Test message (handled by log1)"],"service":"logging-service"}],"msg_services":[]}
D:\dist-sys\Task5>

```

Перевірка збереження повідомлень у logging-service (GET /log)

```

logging-service1-1 | INFO:__main__:[log1] Fetched 1 messages
logging-service1-1 | INFO:werkzeug:172.19.0.1 - - [17/Jun/2025 17:28:54] "GET /log HTTP/1.1" 200 -

```

```

D:\dist-sys\Task5>curl -X GET http://localhost:8881/log
{"messages":["Test message (handled by log1)"],"node":"log1"}

```

Перевірка отримання повідомлень у messages-service (GET /msg)

```

logging-service3-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:29:41] "GET /health HTTP/1.1" 200 -
messages-service1-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:29:43] "GET /health HTTP/1.1" 200 -
messages-service1-1 | INFO:__main__:[msg1] Returning 1 messages
messages-service1-1 | INFO:werkzeug:172.19.0.1 - - [17/Jun/2025 17:29:44] "GET /msg HTTP/1.1" 200 -
facade-service1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:29:47] "GET /health HTTP/1.1" 200 -
logging-service2-1 | INFO:werkzeug:172.19.0.3 - - [17/Jun/2025 17:29:48] "GET /health HTTP/1.1" 200 -

```

```
D:\dist-sys\Task5>curl -X GET http://localhost:8890/msg  
{ "messages": ["Test message"] }
```

Налаштування для клієнтів Hazelcast мають зберігатись як key/value у *Consul* і зчитуватись *logging-service*

config

Search	Type ▾
Name	
hazelcast	
kafka	

hazelcast

Search	Type ▾
Name	
cluster_members	
cluster_name	

cluster_members

Value	
1	["hazelcast1:5701", "hazelcast2:5701", "hazelcast3:5701"]

cluster_name

Value	
1	"dev"

Налаштування для Message Queue (адреса, назва черги, ...) мають зберігатись як key/value у *Consul* і зчитуватись ***facade-service*** та ***messages-service***

< Key / Values / config

kafka

Search	Type ▾
Name	
brokers	
topic	

brokers

Value	
1	["kafka1:9092", "kafka2:9093", "kafka3:9094"]

topic

Value	
1	"msg-queue"

Продемонструвати, що у випадку відключення екземпляру певного мікросервісу, це буде відображатись у Consul (відключений екземпляр сервісу змінить статус) , а виклики будуть перенаправлятись до інших працюючих екземплярів.

```
PS D:\dist-sys\Task5> docker stop task5-logging-service2-1
task5-logging-service2-1
PS D:\dist-sys\Task5>
```

logging-service

Instances	Intentions	Routing	Tags
Q Search Search Across ▼ Health Status ▼			
log2 ✖ All service checks failing ✔ All node checks passing 033f3c47b4f3 logging-service2:8882			
log1 ✔ All service checks passing ✔ All node checks passing 033f3c47b4f3 logging-service1:8881			
log3 ✔ All service checks passing ✔ All node checks passing 033f3c47b4f3 logging-service3:8883			

```
D:\dist-sys\Task5>curl -X POST http://localhost:8880/msg -H "Content-Type: application/json" -d "{\"content\": \"Test message 2\"}"
{"status":"Sent to Kafka and logging-service"}
D:\dist-sys\Task5>
```

```

consul-1 | 2025-06-17T17:33:58.662Z [WARN] agent: Check is now critical: check=service:log2
logging-service1-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:33:59] "GET /health HTTP/1.1" 200 -
messages-service2-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:33:59] "GET /health HTTP/1.1" 200 -
logging-service3-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:01] "GET /health HTTP/1.1" 200 -
facade-service-1 | INFO:._main_: [facade1] Message sent to Kafka: Test message 2
messages-service1-1 | INFO:._main_: [msg1] Consumed message: Test message 2
facade-service-1 | INFO:._main_: [facade1] Fetched URLs for logging-service: ['http://logging-service1:8881', 'http://logging-service2:8882', 'http://lo
gging-service3:8883']
logging-service3-1 | INFO:._main_: [log3] Stored message: Test message 2
facade-service-1 | INFO:._main_: [facade1] Successfully sent to http://logging-service3:8883
logging-service3-1 | INFO:werkzeug:172.19.0.17 -- [17/Jun/2025 17:34:03] "POST /Log HTTP/1.1" 201 -
facade-service-1 | INFO:._main_: [facade1] Successfully sent to logging-service: {'status': 'Stored'}
facade-service-1 | INFO:werkzeug:172.19.0.1 -- [17/Jun/2025 17:34:03] "POST /msg HTTP/1.1" 201 -
messages-service1-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:03] "GET /health HTTP/1.1" 200 -
facade-service-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:09] "GET /health HTTP/1.1" 200 -
logging-service1-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:09] "GET /health HTTP/1.1" 200 -
messages-service2-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:09] "GET /health HTTP/1.1" 200 -
kafka2-1 | [2025-06-17 17:34:09,675] INFO [Controller id=2] Processing automatic preferred replica leader election (kafka.controller.KafkaContro
ller)
kafka2-1 | [2025-06-17 17:34:09,675] TRACE [Controller id=2] Checking need to trigger auto leader balancing (kafka.controller.KafkaController)
kafka2-1 | [2025-06-17 17:34:09,677] DEBUG [Controller id=2] Topics not in preferred replica for broker 1 HashMap() (kafka.controller.KafkaContr
oller)
kafka2-1 | [2025-06-17 17:34:09,677] TRACE [Controller id=2] Leader imbalance ratio for broker 1 is 0.0 (kafka.controller.KafkaController)
kafka2-1 | [2025-06-17 17:34:09,677] DEBUG [Controller id=2] Topics not in preferred replica for broker 2 HashMap() (kafka.controller.KafkaContr
oller)
kafka2-1 | [2025-06-17 17:34:09,677] TRACE [Controller id=2] Leader imbalance ratio for broker 2 is 0.0 (kafka.controller.KafkaController)
kafka2-1 | [2025-06-17 17:34:09,677] DEBUG [Controller id=2] Topics not in preferred replica for broker 3 HashMap() (kafka.controller.KafkaContr
oller)
kafka2-1 | [2025-06-17 17:34:09,677] TRACE [Controller id=2] Leader imbalance ratio for broker 3 is 0.0 (kafka.controller.KafkaController)
consul-1 | 2025-06-17T17:34:10.670Z [WARN] agent: Check is now critical: check=service:log2
logging-service3-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:11] "GET /health HTTP/1.1" 200 -
messages-service1-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:13] "GET /health HTTP/1.1" 200 -
facade-service-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:17] "GET /health HTTP/1.1" 200 -
logging-service1-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:19] "GET /health HTTP/1.1" 200 -
messages-service2-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:19] "GET /health HTTP/1.1" 200 -
logging-service3-1 | INFO:werkzeug:172.19.0.3 -- [17/Jun/2025 17:34:21] "GET /health HTTP/1.1" 200 -
consul-1 | 2025-06-17T17:34:22.672Z [WARN] agent: Check is now critical: check=service:log2

```

```

D:\dist-sys\Task5>curl -X GET http://localhost:8881/log
{"messages":["Test message (handled by log1)","Test message 2 (handled by log3)"],"node":"log1"}

```

```
D:\dist-sys\Task5>
```

Повтор для messages-service

```

PS D:\dist-sys\Task5> docker stop task5-messages-service2-1
task5-messages-service2-1
PS D:\dist-sys\Task5>

```

```

D:\dist-sys\Task5>curl -X GET http://localhost:8880/msg
{"combined_messages":[{"content":"Test message 2"}, {"content":"Test message (handled by log1)"}, {"content":"Test message 2 (handled by log3)"}, {"content":"T
est message"}], "log_services":[{"messages":["Test message (handled by log1)","Test message 2 (handled by log3)","service":"logging-service"}], "msg_services
":["Test message","Test message 2"]}

```